

Identifying Performance Descriptor in Direct Syngas Conversion to Long-chain α -olefins over Ruthenium-based Catalysts Promoted by Alkali Metals

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Experimental Section

Chemicals

Ruthenium nitrosyl nitrate solution (0.1414 g Ru per gram of solution) was purchased from Heraeus Precious Metal Technology Co., Ltd. Aerosol silica (SiO_2 , AEROSIL 380) was purchased from Evonik Degussa China Co., Ltd. Alkali metal nitrates including LiNO_3 , NaNO_3 , KNO_3 , RbNO_3 and CsNO_3 (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd. All materials were used as received without further purification.

Catalyst Preparation

All alkali metal promoted Ru/ SiO_2 catalysts were prepared via incipient wetness impregnation method. The mixture solution of ruthenium nitrosyl nitrate and alkali metal nitrate (molar ratio of alkali metal to Ru = 0.5) was used for the incipient wetness impregnation of aerosol silica (SiO_2). The as-prepared samples denoted as AM-Ru/ SiO_2 (AM = Li, Na, K, Rb, Cs) were dried at 60 °C for 4 h and calcinated in air at 400 °C for 3 h with a ramping rate of 2 °C/min. ICP-OES and XRF results showed the analogical Ru loading (4.0 wt.%) and AM/Ru molar ratio (0.5) for each sample. The promoted catalysts with different K content were prepared via similar procedure with different amount of KNO_3 . The samples were denoted as xK-Ru/ SiO_2 (x=0.1, 0.3, 0.5 and 1.0), where x represents the K/Ru molar ratio.

Catalyst Characterization

The elemental composition of various samples was measured by inductively coupled plasma optical emission spectrometry (ICP-OES, Perkin Elmer) and X-ray fluorescence (XRF, Rigaku ZSX Primus II).

N_2 physisorption experiments were carried out on Micromeritics 2020 instrument at -196 °C. Brunauer-Emmett-Tellwer (BET) method was used to

calculate the specific surface areas. Pore volume and pore size were determined by the Barrett-Joyner-Halenda (BJH) method.

Power X-ray diffraction (XRD) was performed on a Rigaku Ultima IV X-ray diffractometer (40 kV, 40 mA) equipped with Cu K α radiation ($\lambda = 1.54056 \text{ \AA}$), which was operated at 40 kV and 40 mA with a scanning rate of 2 degree/min.

Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) measurements were conducted on a FEI Tecnai G2 F20 S-TWIN equipment with 200 kV accelerating voltage. The nanoparticle size distribution for each sample was determined using at least 200 nanoparticles.

X-ray photoelectron spectroscopy (XPS) data was collected by Thermo Fisher Scientific K-Alpha spectrometer equipped with the Al K α (1486.6 eV) radiation source. The results were calibrated with C1s peak at 284.8 eV.

H₂ temperature-programmed reduction (H₂-TPR) was performed on Micromeritics ChemiSorb2920 instrument with a thermal conductivity detector (TCD). Before TPR measurement, the samples were pretreated in He flow at 120 °C for 60 min, then cooled down to 50 °C. The reactor was heated to 800 °C at a heating rate of 10 °C·min⁻¹ in 5 vol% H₂ (balance Ar, 30 mL·min⁻¹).

H₂ and CO chemisorption experiments were conducted with a Micromeritics 2020 instrument. Typically, 100 to 150 mg of sample was degassed at 300 °C for 4 h. Prior to the chemisorption experiments, the sample was in situ treated at 450 °C in H₂ for 2 h and then evacuated. The temperature was decreased to 40 °C under vacuum, and then was evacuated for 60 min. The adsorbate (H₂ or CO) was introduced into system to record the chemisorption isotherms. In order to compare the effect of adsorbed CO on H₂ adsorption, H₂ chemisorption was also measured over the reduced sample after CO pre-adsorption and evacuation at 40 °C.

In situ diffuse reflectance infrared Fourier transform spectra of CO adsorption (CO-DRIFTS) were conducted on ThermoFisher Scientific FTIR spectrometer (Nicolet iS50). Prior to CO adsorption, the sample was pretreated

in H₂ flow at 450 °C for 1 h and then the gas flow switched to Ar flow to purge and cooled down to 50 °C or 270 °C. The background spectra were collected at 50 °C and 270 °C under Ar flow. After that, the CO flow was introduced until saturated adsorption. The CO-DRIFTS spectra were obtained until no gas-phase CO with Ar purging. As for the adsorbed CO reactivity experiment at 270 °C, a flow of diluted H₂ was introduced into the reactors and the spectra were automatically recorded to acquire the reactivity of adsorbed CO.

Temperature-programmed surface reaction (TPSR) of CO was carried out in a Micromeritics Chemsorb2920 with a thermal conductivity detector (TCD) and MKS Cirrus 2 mass spectrometer (MS). The sample was pretreated in a H₂ flow of H₂ at 450 °C for 2 h and then flushing with He for 30 min. The temperature was cooled down to 50 °C and CO flow was switched to realize the saturated CO adsorption. The sample was heated up to 500 °C at 10 °C·min⁻¹ in flowing of H₂ flow, and the signals of m/z=16 (CH₄) was monitored by mass spectrometer.

X-ray absorption spectroscopy (XAS) experiments were conducted on the BL11B beamline of Shanghai Synchrotron Radiation Facility (SSRF), China. The Ru K-edge spectra were collected at room temperature with transition mode.

Catalytic Evaluation

Syngas conversion over various catalysts was carried out in a continuous fixed bed made of stainless steel. Before catalytic testing, 1 g sample (40–60 mesh) was in situ reduced at 450 °C for 4 h in pure H₂ flowing with a heating ramp of 2 °C/min. The feed gas (H₂/CO=2, including 3% N₂ as inner standard) was introduced after cooling down to 180 °C, then the reaction pressure was controlled at 5 bar. The reaction temperature was adjusted to 270 °C with weight hourly space velocity (WHSV) of 3000 mL·g⁻¹·h⁻¹. The outlet gas compositions were analyzed by an on-line gas chromatograph (Agilent 7890B)

equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). The liquid product was collected and analyzed by off-line Shimadzu GC equipped with FID. In order to obtain similar CO conversion level (< 10%) for various catalysts, the evaluation was performed by adjusting reaction space velocity.

CO conversion was calculated according to the following equation:

$$X_{CO} = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} \times 100\% \quad (1)$$

Where CO_{inlet} and CO_{outlet} refer to the molar numbers of CO at the inlet and outlet, respectively.

Product selectivity was calculated according to the following equation:

$$S_i = \frac{N_i \times n_i}{\sum(N_i \times n_i)} \times 100\% \quad (2)$$

where S_i denotes the selectivity to product i on a carbon basis, N_i is the molar fraction of product i, and n_i is the carbon number of product i. The selectivity of oxygenates was less than 1% and has been excluded from the reported product selectivity.

The reaction rate was calculated as following:

$$\text{Reaction rate}_{CO} = \frac{WHSV \times X_{CO} \times CO \text{ Concentration}}{22400 \times \xi_{Ru}} \quad (3)$$

Where WHSV is the weight hourly space velocity, X_{CO} is the CO conversion, CO Concentration represents the molar concentration of CO in the feedstock, ξ_{Ru} represents the loading amount of Ru measured by ICP-AES.

The turnover frequency (TOF) value was calculated as following:

$$TOF = \frac{\text{Reaction rate}_{CO} \times M_{Ru}}{D_{Ru} \times 3600} \quad (4)$$

Where M_{Ru} is the relative atomic mass of Ru (101.07 g/mol), D_{Ru} is the dispersion of metallic Ru measured by CO chemisorption.

The chain growth probability (α) was calculated according to Anderson-Schulz-Flory distribution:

$$\left(\frac{W_n}{n}\right) = (1 - \alpha)^2 \alpha^{(n-1)} \quad (5)$$

where n is the carbon number of products, W_n is mass fraction of the hydrocarbons with a carbon number of n , α is chain growth probability.

The equation of (4) can be edited as:

$$\ln\left(\frac{W_n}{n}\right) = (n - 1)\ln\alpha + 2\ln(1 - \alpha) \quad (6)$$

Plotting $\ln(W_n/n)$ versus n (carbon number), and the chain growth probability (α) can be obtained by calculating the slope ($\ln\alpha$).

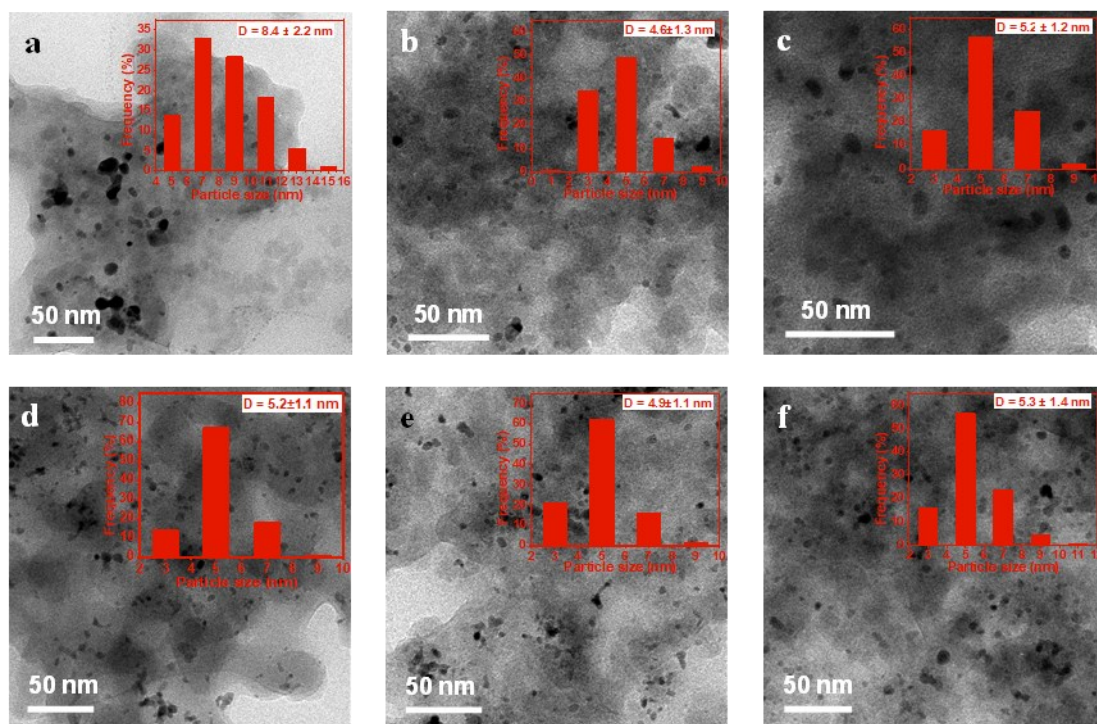


Figure S1. Representative TEM images and the corresponding crystallite size distributions for various spent catalysts. (a) Ru/SiO₂, (b) Li-Ru/SiO₂, (c) Na-Ru/SiO₂, (d) K-Ru/SiO₂, (e) Rb-Ru/SiO₂, (f) Cs-Ru/SiO₂.

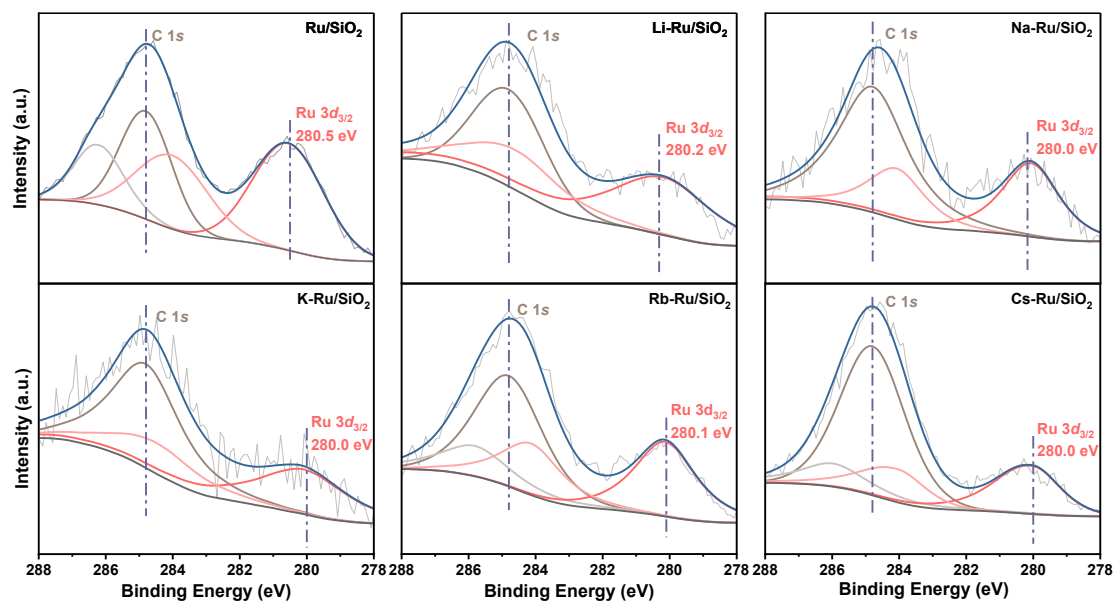


Figure S2. XPS Ru 3d spectra of various AM-Ru/SiO₂ catalysts after reduction by H₂ at 450 °C for 4 h.

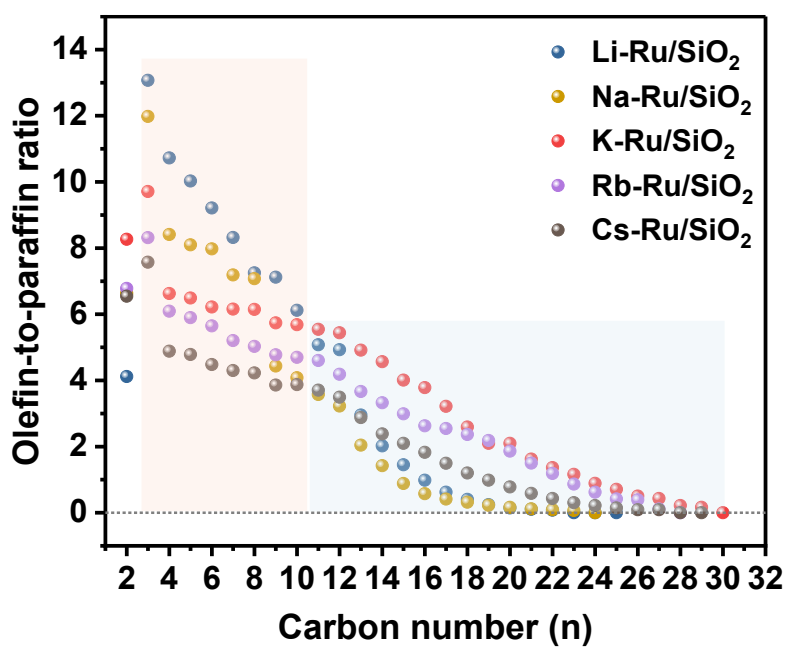


Figure S3. Effect of alkali metal promoter on the olefin-to-paraffin ratio of hydrocarbons.

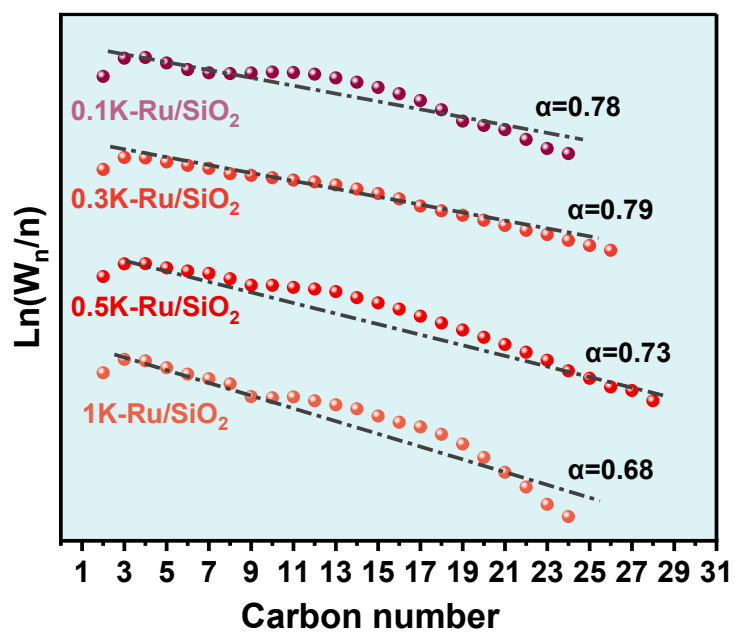


Figure S4. ASF distribution of olefins products for various xK-Ru/SiO₂ catalysts at 270 °C, 5 bar, 3000 mL·g_{cat.}⁻¹·h⁻¹ and H₂/CO=2.

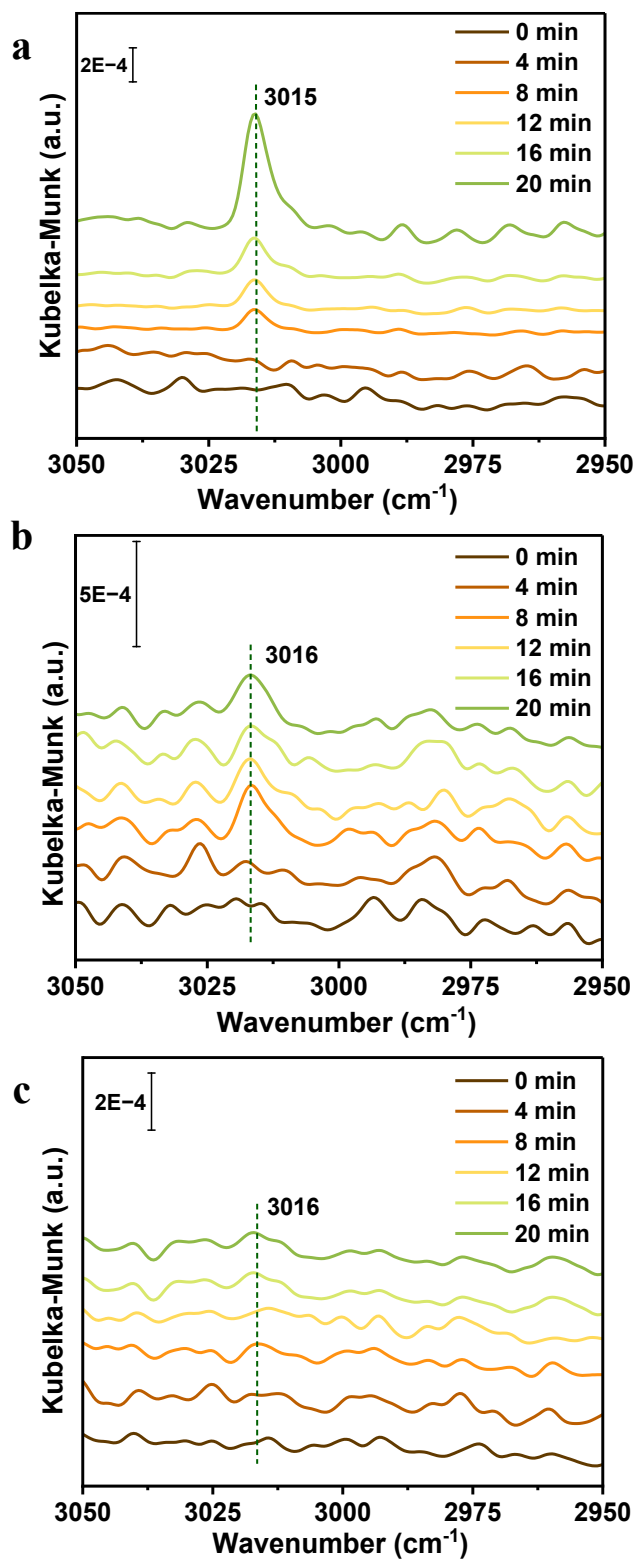


Figure S5. Generation of CH_x species during H_2 flow at 270°C as determined using in situ CO-DRIFTs over (a) Li-Ru/ SiO_2 , (b) K-Ru/ SiO_2 and (c) Cs-Ru/ SiO_2 .

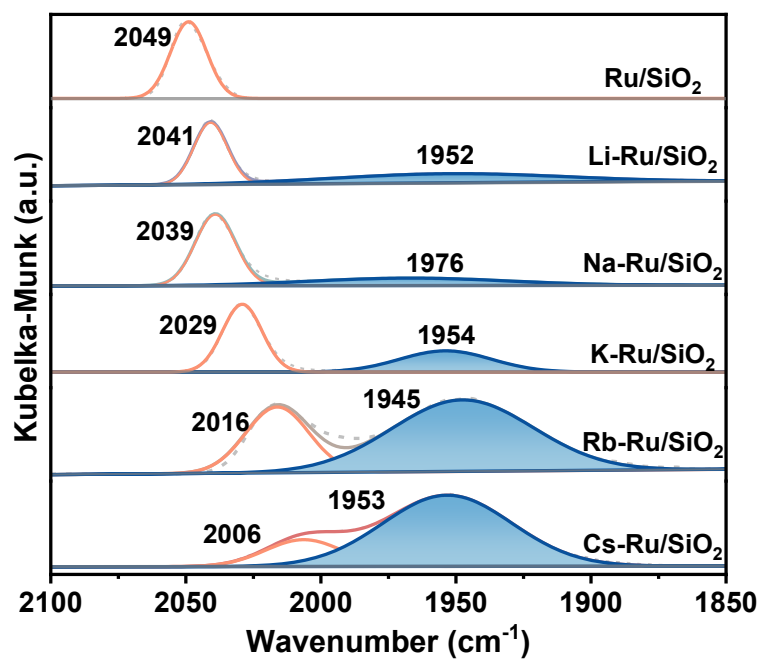


Figure S6. CO-DRIFTS spectra with partial enlarged detail at the range of 2100 –1850 cm⁻¹ over AM-Ru/SiO₂ catalysts.

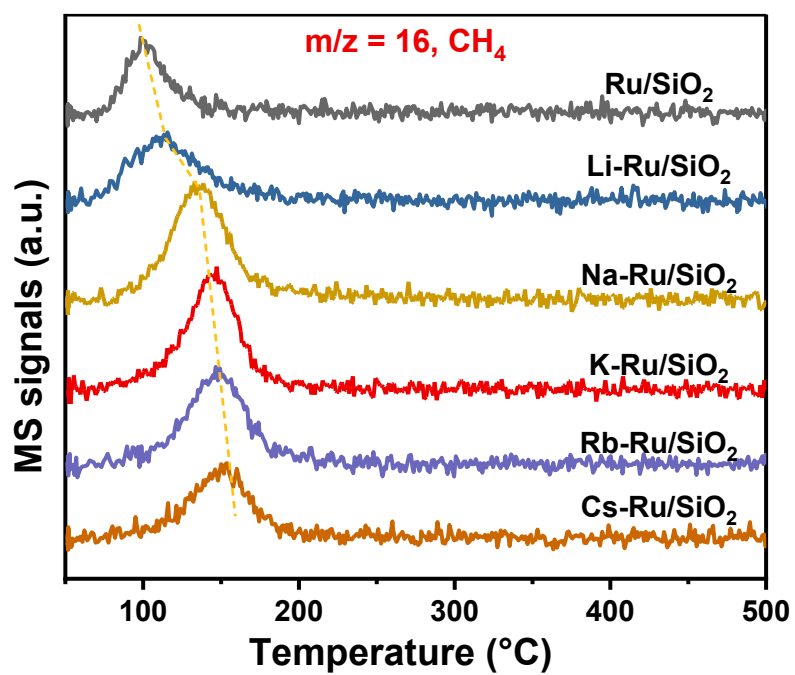


Figure S7. CO-TPSR profiles for AM-Ru/SiO₂ catalysts.

Table S1. EXAFS fitting parameters at the Ru-edge ($S_0^2=0.75$).

Sample	Shell	CN ^a	^b R(Å)	^c $\sigma^2 \times 10^2 (\text{\AA}^2)$	^d $\Delta E_0 (\text{eV})$	R factor
Ru foil	Ru-Ru	12.0	2.71	-	-	-
Li-Ru/SiO ₂	Ru-Ru	10.9	2.67	0.42±0.05	6.27±1.19	0.007
Na-Ru/SiO ₂	Ru-Ru	10.5	2.67	0.46±0.05	5.55±1.06	0.006
K-Ru/SiO ₂	Ru-Ru	10.3	2.67	0.41±0.05	5.47±1.05	0.009
Rb-Ru/SiO ₂	Ru-Ru	10.0	2.67	0.44±0.05	5.05±1.05	0.006
Cs-Ru/SiO ₂	Ru-Ru	10.7	2.67	0.45±0.07	5.66±1.34	0.008
RuO ₂	Ru-O	6.0	1.99	-	-	-

^a CN is the coordination number for the adsorber-backscatter pair.

^b R is the average interatomic distance.

^c σ^2 is the Debye-Waller factor.

^d ΔE_0 is the inner potential correction.

Table S2. Chemisorption results of various AM-Ru/SiO₂ catalysts.

Catalysts	H/Ru	H/Ru ^a	CO/Ru
Li-Ru/SiO ₂	0.14	0.11	0.30
Na-Ru/SiO ₂	0.16	0.09	0.31
K-Ru/SiO ₂	0.26	0.14	0.33
Rb-Ru/SiO ₂	0.20	-	0.31
Cs-Ru/SiO ₂	0.16	0.12	0.30

^a The H/Ru value was calculated after CO pre-adsorbed.